American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
6	11/5/27	9:00 am	50 mins	◉ Advisor ○ Students	Raghad Al Agha

Attendees:

Raghad Al Agha, Lynn Ariss, Hadi Elham, Claritta Khoury

Briefly summarize the main discussions during the meeting:

The meeting focused on finalizing the heatmap sensing circuit and resolving hardware power challenges while continuing software and app development. The pressure matrix (27×11 grid with 5 mm copper pads and 5 mm spacing) is now functioning, using three multiplexers—two for digital column scanning and one for analog row reading—operating at microsecond-level switching speeds. Data is streamed to Python for interpolation, smoothing, and visualization. The team discussed matrix resolution limits, normalization, and improving spatial accuracy. Power delivery was a major concern: the Arduino requires stable 4.5 – 5V input for proper regulation to 3.3V, but the system currently relies on a 3.7V Li-Po battery. Options discussed included using a step-up converter (3.75V), bypassing the onboard regulator and supplying regulated 3.3V directly, or redesigning around double batteries instead of one then using a step down converter. App updates include corrected timestamp logging in CSV exports and implementation of random key generation for encryption, though Bluetooth pairing and key persistence still present inconsistencies. On the signal-processing side, model tuning continues with 5-second windows, while PPG algorithm robustness remains an issue due to noisy peak detection and unreliable HR/SpO readings. The team is experimenting with alternative open-source PPG libraries and interrupt handling approaches.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

The pressure matrix hardware is operational and sufficiently fast, but resolution optimization and stable normalization remain ongoing challenging tasks. A critical hardware priority is implementing a reliable power solution—likely via a DC-DC step-up converter to 5V or a carefully regulated 3.3V bypass—ensuring stable operation without damaging the board. App functionality (CSV logging, timestamps, encryption) is progressing, but Bluetooth-key management and PPG signal reliability require further debugging. The current focus is on stabilizing voltage regulation, improving PPG robustness, and validating consistent heart-rate readings before advancing to higher-level model refinements or clinical testing. Achieving clinically approved ranges and values for classifying water retention/swelling is a challenge to be tackled as well.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Debug the peak detection algorithm to ensure it works and to test it against our already existing one, write conference paper	Claritta Khoury	Wednesday March 4
Data processing for heatmap generation and reconstruction for better accuracy, and also potentially integrating iot on the app, write conference paper	Hadi Elham	Wednesday March 4
Finalizing flex sensor algorithm, order components, write conference paper, reconstruction of heatmap for better resolution	Raghad Al Agha	Wednesday March 4
	Lynn Ariss	Wednesday March 4